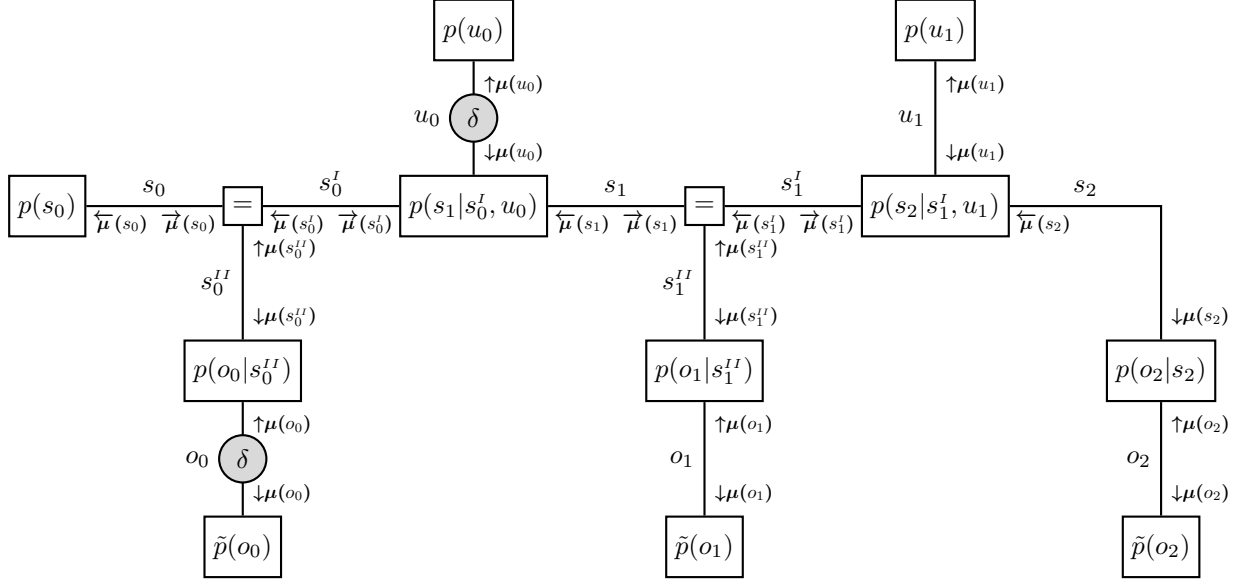


Setup BFE (current timestep $t = 1$)



Bethe Forward Sweep:

$$\vec{\mu}(s_0) = p(s_0) \quad (1)$$

$$\uparrow\mu(o_0) = \delta_{o_0, \hat{o}_0} \quad (2)$$

$$\uparrow\mu(s_0^{II}) = \exp\left(\sum_{o_0} q(o_0) \log p(o_0|s_0^{II})\right) \quad (3)$$

$$\vec{\mu}(s_0^I) = \vec{\mu}(s_0) \uparrow\mu(s_0^{II}) \quad (4)$$

$$\downarrow\mu(u_0) = \delta_{u_0, \hat{u}_0} \quad (5)$$

$$\vec{\mu}(s_1) = \sum_{s_0^I, u_0} \vec{\mu}(s_0^I) \downarrow\mu(u_0) p(s_1|s_0^I, u_0) \quad (6)$$

$$\uparrow\mu(o_1) = \tilde{p}(o_1) \quad (7)$$

$$\uparrow\mu(s_1^{II}) = \exp\left(\sum_{o_1} q(o_1) \log p(o_1|s_1^{II})\right) \quad (8)$$

$$\vec{\mu}(s_1^I) = \vec{\mu}(s_1) \uparrow\mu(s_1^{II}) \quad (9)$$

$$\downarrow\mu(u_1) = p(u_1) \quad (10)$$

$$\downarrow\mu(s_2) = \sum_{s_1^I, u_1} \vec{\mu}(s_1^I) \downarrow\mu(u_1) p(s_2|s_1^I, u_1) \quad (11)$$

$$\downarrow\mu(o_2) = \exp\left(\sum_{s_2} q(s_2) \log p(o_2|s_2)\right) \quad (12)$$

Bethe Backward Sweep:

$$\uparrow\mu(o_2) = \tilde{p}(o_2) \quad (13)$$

$$\overleftarrow{\mu}(s_2) = \exp\left(\sum_{o_2} q(o_2) \log p(o_2|s_2)\right) \quad (14)$$

$$\uparrow\mu(u_1) = \sum_{s_1^I, s_2} \vec{\mu}(s_1^I) \overleftarrow{\mu}(s_2) p(s_2|s_1^I, s_2) \quad (15)$$

$$\overleftarrow{\mu}(s_1^I) = \sum_{u_1, s_2} \downarrow\mu(u_1) \overleftarrow{\mu}(s_2) p(s_2|s_1^I, u_1) \quad (16)$$

$$\downarrow\mu(s_1^{II}) = \vec{\mu}(s_1) \overleftarrow{\mu}(s_1^I) \quad (17)$$

$$\downarrow\mu(o_1) = \exp\left(\sum_{s_1^{II}} q(s_1^{II}) \log p(o_1|s_1^{II})\right) \quad (18)$$

$$\overleftarrow{\mu}(s_1) = \uparrow\mu(s_1^{II}) \overleftarrow{\mu}(s_1^I) \quad (19)$$

$$\uparrow\mu(u_0) = \delta_{u_0, \hat{u}_0} \quad (20)$$

$$\overleftarrow{\mu}(s_0^I) = \sum_{u_0, s_1} \downarrow\mu(u_0) \overleftarrow{\mu}(s_1) p(s_1|s_0^I, u_0) \quad (21)$$

$$\downarrow\mu(s_0^{II}) = \vec{\mu}(s_0) \overleftarrow{\mu}(s_0^I) \quad (22)$$

$$\downarrow\mu(o_0) = \delta_{o_0, \hat{o}_0} \quad (23)$$

$$\overleftarrow{\mu}(s_0) = \uparrow\mu(s_0^{II}) \overleftarrow{\mu}(s_0^I) \quad (24)$$

Updating Beliefs:

$$q(s_0) \propto \overleftarrow{\mu}(s_0) \vec{\mu}(s_0) \quad (25)$$

$$q(s_0^I) \propto \overleftarrow{\mu}(s_0^I) \vec{\mu}(s_0^I) \quad (26)$$

$$q(s_0^{II}) \propto \uparrow\mu(s_0^{II}) \downarrow\mu(s_0^{II}) \quad (27)$$

$$q(s_1) \propto \overleftarrow{\mu}(s_1) \vec{\mu}(s_1) \quad (28)$$

$$q(s_1^I) \propto \overleftarrow{\mu}(s_1^I) \vec{\mu}(s_1^I) \quad (29)$$

$$q(s_1^{II}) \propto \uparrow\mu(s_1^{II}) \downarrow\mu(s_1^{II}) \quad (30)$$

$$q(s_2) \propto \overleftarrow{\mu}(s_2) \downarrow\mu(s_2) \quad (31)$$

$$q(u_0) \propto \delta_{u_0, \hat{u}_0} \quad (32)$$

$$q(u_1) \propto \uparrow\mu(u_1) \downarrow\mu(u_1) \quad (33)$$

$$q(o_0) \propto \delta_{o_0, \hat{o}_0} \quad (34)$$

$$q(o_1) \propto \uparrow\mu(o_1) \downarrow\mu(o_1) \quad (35)$$

$$q(o_2) \propto \uparrow\mu(o_2) \downarrow\mu(o_2) \quad (36)$$

$$\begin{aligned}
F = & \sum_{s_0} q(s_0) \log \frac{q(s_0)}{p(s_0)} + \sum_{s_1, s_0^I, u_0} q(s_1, s_0^I, u_0) \log \frac{q(s_1, s_0^I, u_0)}{p(s_1|s_0^I, u_0)} + \sum_{s_2, s_1^I, u_1} q(s_2, s_1^I, u_1) \log \frac{q(s_2, s_1^I, u_1)}{p(s_2|s_1^I, u_1)} \\
& + \sum_{u_0} q(u_0) \log \frac{q(u_0)}{p(u_0)} + \sum_{u_1} q(u_1) \log \frac{q(u_1)}{p(u_1)} \\
& + \sum_{o_0, s_0^{II}} q(o_0, s_0^{II}) \log \frac{q(o_0, s_0^{II})}{p(o_0|s_0^{II})} + \sum_{o_1, s_1^{II}} q(o_1, s_1^{II}) \log \frac{q(o_1, s_1^{II})}{p(o_1|s_1^{II})} + \sum_{o_2, s_2} q(o_2, s_2) \log \frac{q(o_2, s_2)}{p(o_2|s_2)} \\
& + \sum_{o_0} q(o_0) \log \frac{q(o_0)}{\tilde{p}(o_0)} + \sum_{o_1} q(o_1) \log \frac{q(o_1)}{\tilde{p}(o_1)} + \sum_{o_2} q(o_2) \log \frac{q(o_2)}{\tilde{p}(o_2)} \\
& + \sum_{s_0, s_0^I, s_0^{II}} q(s_0, s_0^I, s_0^{II}) \log \frac{q(s_0, s_0^I, s_0^{II})}{\delta_{s_0, s_0^I} \delta_{s_0, s_0^{II}}} + \sum_{s_1, s_1^I, s_1^{II}} q(s_1, s_1^I, s_1^{II}) \log \frac{q(s_1, s_1^I, s_1^{II})}{\delta_{s_1, s_1^I} \delta_{s_1, s_1^{II}}} \\
& - \sum_{s_0} q(s_0) \log q(s_0) - \sum_{s_0^I} q(s_0^I) \log q(s_0^I) - \sum_{s_0^{II}} q(s_0^{II}) \log q(s_0^{II}) \\
& - \sum_{s_1} q(s_1) \log q(s_1) - \sum_{s_1^I} q(s_1^I) \log q(s_1^I) - \sum_{s_1^{II}} q(s_1^{II}) \log q(s_1^{II}) \\
& - \sum_{u_0} q(u_0) \log q(u_0) - \sum_{u_1} q(u_1) \log q(u_1) \\
& - \sum_{o_0} q(o_0) \log q(o_0) - \sum_{o_1} q(o_1) \log q(o_1) - \sum_{o_2} q(o_2) \log q(o_2)
\end{aligned}$$